

- ☒ closeness to roads and hedges metric.
 - ☒ define edges on map—how?
- ☐ altitude and speed plots.
- ☐ group trajectories, statistics of trajectories.
 - grids. time spent in grids, distribution.
- ☒ clustering
- ☐ MSE of each pigeon from a given release site vs optimal.
 - ☐ see if the MSEs cluster.
- ☐ altitude—where do they perch? what are the weather conditions?
 - ☐ when speed = 0, not at the tail end and altitude constant. What is the lat/lon? What does it look like?
 - ☐ what are the weather conditions at that time?

guiding principles for pigeon flight

- ☐ follow roads
 - ☐ any metric to determine this?
 - ☐ findmin distance to a road, any road (computationally expensive)
 - ☐ sum all min distances.
 - ☐ label all road segments and return labels as well as distances as a tuple.
 - ☐ how?
 - ☐ radius of vision.
 - ☐ dependent on weather/altitude?
 - ☐ where pigeons are between roads (high min distance to nearby road), correlate with weather and altitude. does pigeon adjust bearing?
 - ☐ initial bearing from sun/EM
 - ☐ if yes, then in initial (low-learning) scenarios, cities may confuse birds. extreme trajectories statistics correlated with area.
 - ☐ mark density of roads on map. probability of landing in bounding box of areas?
 - ☐ thickness of trajectories, and area of trajectories inside bounding box.
 - ☐ does shape of bounding box matter?

- learning?
 - clustering: MSE
 - decrease in spread: MSE
 - increased frequency of previously used paths: histograms of time spent in grids, make sure shape of grids does not matter.
 - more or less reliance on roads: MSE, min distance from roads statistics.
 - individual pigeon's usage of previous trajectories/roads?
- flow: data—hypothesis—model—verify on data (example use R1, R2 to build and R3 to verify, permute)—revise model—possibly general inference about learning dynamics.